When a solid is added to a solvent, it begins to dissolve. i.e. the solute leaves from the solid state (dissolution). After some time, some of the dissolved solute returns back to the solid state (recrystallisation). If there is excess of solid present, the rate of both these processes becomes equal at a particular stage. At this stage an equilibrium is established between the solid solute molecules and dissolved solute molecules.

$$\text{Solute (solid)} \leftrightarrows \text{Solute (dissolved)}$$

According to Le-Chatelier principle, if the dissolution process is endothermic, the increase in temperature will shift the equilibrium towards right i.e solubility increases. for an exothermic reaction, the increase in temperature decreases the solubility. The solubilities of ammonium nitrate, calcium chloride, ceric sulphate nano-hydrate and sodium chloride in water at different temperatures are given in the following graph.

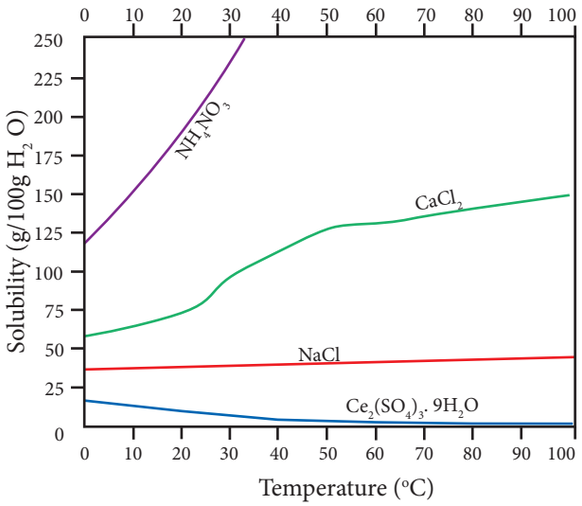


Figure 9. 1 Plot of solubility versus temperature for selective compounds

The following conclusions are drawn from the above graph.

- The solubility of sodium chloride does not vary appreciable as the maximum solubility is achieved at normal temperature. In fact, there is only 10 % increase in solubility between 0 ˚ to 100 ˚C.
- The dissolution process of ammonium nitrate is endothermic, the solubility increases steeply with increase in temperature.
- In the case of ceric sulphate, the dissolution is exothermic and the solubility decreases with increase in temperature.
- Even though the dissolution of calcium chloride is exothermic, the solubility increases moderately with increase in temperature. Here, the entropy factor also plays a significant role in deciding the position of the equilibrium.

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